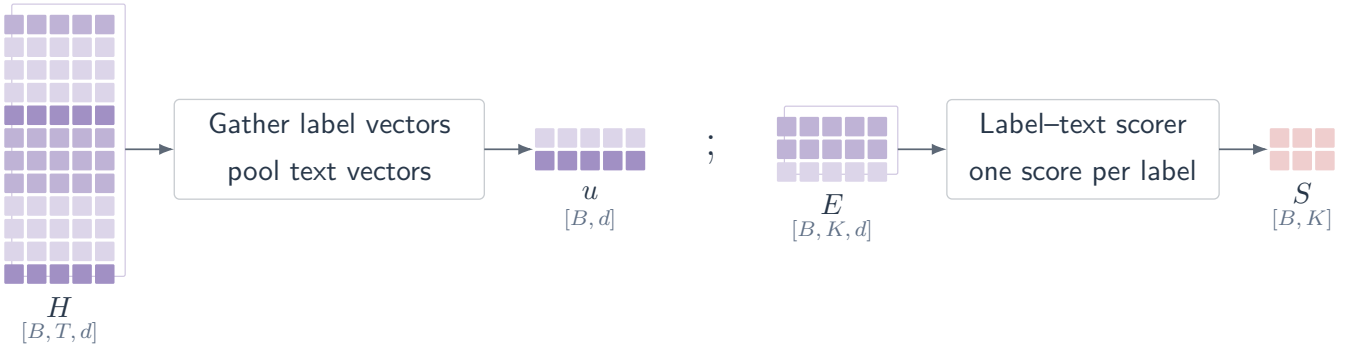
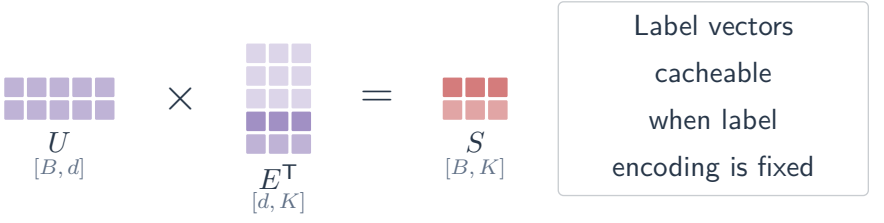
$$S^{uni}_{b,k} = g(H^{label}_{b,k}, \mathrm{pool}(H^{text}_{b})), \quad S^{bi} = UE^{\mathsf{T}}/\tau, \quad S^{cross}_{r} = f(Encoder(x_b, c_k))$$

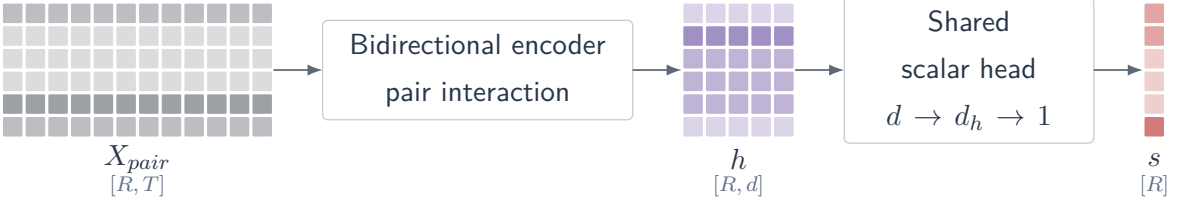
Uni-encoder: encode labels and text together, then compare representations



Bi-encoder: independent encoders produce a matrix of pairwise scores



Cross-encoder: jointly encode each independent text–candidate pair



Axes $U : [B, d]$ holds text vectors; label vectors have shape $E : [B, K, d]$ (uni) or $[K, d]$ (bi). $R = \sum_b K_b$ counts cross-encoder rows; τ is score temperature.

Meaning Visibility depends on the layout. Uni-encoders permit label interaction; bi-encoders separate label encoding; cross-encoders expose text and one candidate to each other. There is no hardware-independent latency or accuracy ordering.

Cell counts illustrate axes, not model sizes; tint identifies roles, not numeric values.